Zahra Cheraghi

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Education

BS **University of Isfahan**, Computer Science — Sept 2022 - Sept 2026

- **GPA 17.29/20** (Faculty Average 14.45/20)
- **Last Year GPA 19.85/20**
- Relevant Coursework: Data Mining (20/20), Artificial Intelligence (19.95/20), Machine Learning (19/20), Computer Networks (20/20), DataBase (18.5/20), Theory of Computation (19.5/20), Graph Theory (18/20), Non-Linear Optimization (19.75/20) , Theory of Coding (17.2/20)

Research Interests

- **Autonomous Systems and ML:** Implementing Reinforcement Learning and Neural Networks to enable self-optimizing and adaptive behavior in complex environments.
- **Evolutionary Computation:** Developing Genetic Algorithms to solve high-dimensional optimization problems and multi-objective design challenges in complex systems.
- **Bioinformatics:** Sequence Analysis and statistical modeling of biological data.
- **Network Algorithm and Optimization:** Designing efficient routing and load-balancing algorithms to minimize latency and maximize throughput in high-speed distributed systems.

Research

Neural Ranking of Perturbed Biological Pathways — GitHub

Supervisor: Dr. Fatemeh Mansoori

- Built an R/`limma` pipeline for gene-level differential-expression analysis of 18 GEO microarray datasets and sample-resampled subsets of 6, 12, 24, and 48 observations.
- Developed a PyTorch neural network with a RankNet-style architecture to rank perturbed biological pathways from expression profiles spanning 22,143 genes and 320 KEGG pathways, without ground-truth pathway labels.
- Investigated contrastive learning algorithms and loss functions inspired by Bootstrap Your Own Latent (BYOL) for training neural networks, focusing on ranking consistency between full and reduced datasets and discrimination between biological conditions.
- Used study-disjoint five-fold cross-validation and a held-out test set; achieved 77.6% top-20 recall and 90.7% discrimination, with 84.4% top-50 recall and 100% top-50 discrimination.

Experience

Teaching Assistant, Machine Learning — Isfahan, IR

Sept 2026 – Present

GitHub

- The course covered supervised and unsupervised learning, including regression, classification, optimization, and model evaluation.
- Taught in-class sessions on machine learning concepts.
- Led hands-on sessions on machine learning models.
- Prepared exercises to reinforce course concepts.
- Prepared course exams.

Teaching Assistant, Database — Isfahan, IR
Sept 2026 – Present
GitHub

- The course covered relational databases, including schema design, SQL queries and joins, constraints, and normalization.
- Prepared exercises on database concepts and SQL.
- Graded student assignments.
- Prepared course exams.

Frontend Developer, PayaTech / Dunijet — Isfahan, IR
June 2025 – Sep 2025

- Built responsive dashboards and real-time sensor interfaces using React/Next.js, interactive charts and tables, and Socket.IO.
- Implemented reusable components, integrated REST/WebSocket APIs, and refined UX with product and design teams.

Projects

Graph Search in a Dynamic Grid World — GitHub

- Implemented BFS, DLS/IDS, UCS, and A* for a multi-target environment with weighted terrain, enemies, weapons, and time-dependent state.
- Designed a cached Dijkstra/MST-based A* heuristic that preserved UCS's optimal path cost while reducing expanded states from 320,721 to 394 on the medium benchmark.

Sensor Placement with Local Search — GitHub

- Implemented hill climbing with restarts, simulated annealing, a genetic algorithm, and beam search to optimize sensor placement across obstacle-filled grid maps.
- Created a shared objective balancing target coverage, sensor count, and coverage overlap, then benchmarked all four methods reproducibly across seven maps.

Adversarial Othello Agent — GitHub

- Built Minimax and Alpha-Beta agents with move ordering, exact endgame search, and a stage-aware evaluation function using mobility, corners, edges, frontier discs, and positional risk.
- Alpha-Beta matched Minimax decisions while examining up to 15.16$\times$ fewer nodes and running up to 11.88$\times$ faster; achieved 100% wins in 20-game tests against random and greedy agents.

Constraint Satisfaction Solvers — GitHub

- Modeled exam scheduling and a 6×6 Sudoku as CSPs using backtracking, MRV, degree heuristics, forward checking, and AC-3 constraint propagation.
- Reduced Sudoku search from 204 generated nodes with plain backtracking to 35 with MRV and forward checking, while also enumerating and optimizing valid exam schedules.

Logic-Based Minesweeper Agent — GitHub

- Built a Python/PyGame agent with PyDatalog safety and danger rules, frontier subset inference, and a risk-based fallback for ambiguous states.
- Solved deterministic 9×9, 15×15, and 45×80 benchmarks to 100% completion, including a 3,600-cell board in 8.4 seconds.

Theory of Coding — GitHub

- Developed a comprehensive simulation of a noisy communication channel integrating Huffman Compression for source coding and Hamming (n, k) Error Correction for channel coding.
- Implemented a noise injection model to simulate bit-flip probabilities (p) and evaluated the system's robustness using Bit Error Rate (BER) analysis.

- Conducted performance benchmarking against the Shannon Limit, visualizing the trade-offs between compression efficiency, error-correction overhead, and the theoretical channel capacity threshold.

Machine Learning GitHub

- Developed a comprehensive library of core Machine Learning models using Python and Jupyter Notebooks, focusing on both supervised and unsupervised learning
- Implemented and optimized various architectures, including Logistic Regression for classification, Neural Networks for pattern recognition, and Support Vector Machines (SVM).
- Designed robust systems for Anomaly Detection and high-dimensional data clustering, ensuring model scalability and mathematical accuracy in algorithm derivation.

Network Project GitHub

- Contributed to a team-based development of an automated object detection system designed to identify and categorize specific items within image and video streams.
- Collaborated on the integration of deep learning models (e.g., YOLO or SSD) to achieve real-time inference and high classification accuracy across diverse environmental conditions.

University Presentations

Support Vector Machines (SVM) GitHub

- Presented support vector machines at university, explaining binary classification, optimal separating hyperplanes, and margin maximization.
- Explained soft-margin optimization and the role of slack variables in balancing classification errors and model generalization.
- Implemented an SVM classifier and demonstrated how soft-margin parameter tuning affects decision boundaries for noisy data and outliers.

Genetic Algorithms GitHub

- Presented genetic algorithms at university, explaining population-based search, fitness evaluation, selection, and mutation.
- Developed and presented a genetic algorithm for the 0/1 knapsack problem, demonstrating fitness scoring and penalties for solutions that exceed weight limits.
- Compared tournament and roulette-wheel selection and explained how mutation rates balance exploration and exploitation.

Certificates

Advanced Python Programming and Object-Oriented Thinking October 2026
Quera College Certificate

- Completed the course with a perfect score.

Technologies

Languages: JavaScript, Python, C, CSS, HTML, React, R

Skills & Technologies: Numpy, Pandas, Matplotlib, Scikit-Learn, Pytorch, Git, Github, SPSS

Language Proficiency

Persian (Farsi) Native proficiency

English TOEFL score: 100